Ornithology Week 8 Practice Problems: Migration

1) Imagine you had a phylogeny of a diverse group of birds. A majority spend both the winter and breeding season in temperate Eurasia. Two small clades within this larger tree spend the breeding season in Eurasia and the winter in Africa. Does this support the hypothesis that seasonal migration evolved as a strategy for reduced competition for nesting sites? Why or why not?

2) Partial migration likely evolved in response to spatially varying selection pressures across a species' range. (**T**, **F**, or **It Depends**)

3) Which of the following offer *indirect* tracking of avian migration? (You may circle more than one answer.)

   a) GPS tags
   b) banding
   c) stable isotopes
   d) genetic data
   e) radio telemetry

4) Define "zugunruhe".

5) How do experiments take advantage of zugunruhe in captive birds to determine how they navigate during migration?

6) Approximately how many bird species are migratory?

   a) 18
   b) 180
   c) 1800
   d) 18000

7) Seasonal migration only evolved once. (**T**, **F**)